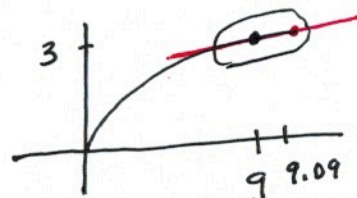


Section 3.10 Linear Approximations

$$\sqrt{9} = 3 \quad \sqrt{9.09} \stackrel{?}{=} 3.03$$

$$f(x) = \sqrt{x}$$



Find tangent line at $(9, 3)$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(9) = \frac{1}{2\sqrt{9}}$$

$$\text{slope} = \frac{1}{6}$$

$$y - 3 = \frac{1}{6}(x - 9)$$

$$y - 3 = \frac{1}{6}x - \frac{9}{6} + 3$$

$$y = \frac{1}{6}x - \frac{3}{2} + \frac{6}{2}$$

$$y = \frac{1}{6}x + \frac{3}{2}$$

$$y = \frac{1}{6}(9.09) + \frac{3}{2}$$

$$y = 3.015$$

Ex (2) Estimate $\sin(3)$ using linearization

$$f(x) = \sin x \quad \text{when } x = \pi$$

$$\sin(\pi) = 0$$

$$\downarrow$$
$$(\pi, 0)$$

$$f'(x) = \cos x$$

$$f'(\pi) = \cos \pi$$

$$\text{Slope} = -1$$

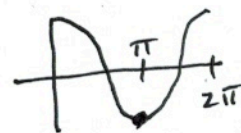
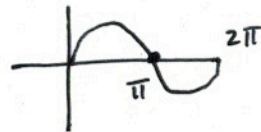
$$y - 0 = -1 \cdot (x - \pi)$$

$$y = -x + \pi$$

$$y = -3 + \pi = -3 + 3.14159 \dots$$

$$= 0.14159 \dots$$

$$\text{Actual Answer : } \sin(3) = 0.14112 \dots$$



$$f(x) \approx f(a) + f'(a)(x-a)$$

Ex ③ Estimate $\sqrt[3]{8.1}$

$$f(x) = \sqrt[3]{x} \quad \text{At } x = 8$$

$$y = \sqrt[3]{8} = 2$$

$$f(x) = x^{1/3}$$

$$f'(x) = \frac{1}{3} x^{-2/3}$$

$$f'(x) = \frac{1}{3 x^{2/3}}$$

$$f'(8) = \frac{1}{3 (8)^{2/3}} = \frac{1}{12}$$

$$f(x) \approx 2 + \frac{1}{12}(x-8)$$

$$f(8.1) \approx 2 + \frac{1}{12}(8.1-8)$$

$$2.0083$$